System Architecture & Technical Specification Document

Sri Lanka Navy — Civil Engineering Department (Eastern Naval Area)
Platform: Naval Civil Works Productivity Suite (NCW-PS / CMSys V2)
Document Code: `CMSYS-SAD-2026-V2.3`
Classification: Official / Internal Engineering Use
Author: Lead Full Stack Developer & Principal System Architect
Date: October 2026

1. Executive Summary & Purpose

The **Civil Engineering Management System (CMSys)** is an integrated enterprise operational and logistics ecosystem developed for the **Civil Engineering Department (Eastern Naval Area), Sri Lanka Navy**. The suite orchestrates human capital (667 naval engineering ratings), statutory vote expenditure, tactical engineering work orders, workshop job cards, multi-zone building material inventory, and facility lifecycle management across base infrastructure and detached naval out-stations.

This document formally captures the complete technical architecture of the system, with special focus on the active repository (Unified Realtime Database Management, Audit & Integrity Portal) and its federated sibling applications within the naval dockyard network.

2. High-Level Architectural Topology

CMSys utilizes a **Federated Hybrid Architecture** combining cloud-hosted real-time document stores (Firebase Realtime Database) with an on-premises relational persistence layer (MySQL via XAMPP) designed for zero-downtime military field operations.

3. Subsystem Breakdown

3.1 CMSys-DB-Hub (Active Repository)

- **Path:**
- **Role:** Centralized Data Governance, Realtime Audit & Schema Inspector.
- **Stack:** Pure HTML5, ES2024 JavaScript, Tailwind CSS (Dark Palette), FontAwesome 6.5.1, JetBrains Mono.
- **Failover Routing:**
    1. *Primary:* `https://cmsys-v2-1fb05-default-rtdb.asia-southeast1.firebasedatabase.app`
    2. *Secondary:* `https://ncw-ps-operations-default-rtdb.asia-southeast1.firebasedatabase.app`
    3. *Local Offline Fallback:* `../CMSys V2/cmsys-v2-1fb05-seed.json`

Primary Functional Modules in CMSys-DB-Hub:

1. **Live Node Explorer:** Real-time data browser for 11 core collections with multi-criteria filtering (`contains` , `equals` , `starts with` , `ends with` , `empty`), multi-column sorting, column hiding, and full JSON payload modal inspector.
2. **Interconnection Matrix (`REL-01` to `REL-26`):** Live map tracking foreign keys, cardinalities, and relationship descriptions across the NoSQL schema graph.
3. **Data Integrity & Audit Validator:** Real-time scanner executing automated sanity checks:
    - Verification of the **Headcount Conservation Law** across 13 trade branches.
    - Orphaned record detector (Job Cards without valid parent Work Orders).
    - Real-time stock valuation aggregator across store categories (BMS, Plumbing, Metal, Tools, Paint).
4. **Snapshot Backup & Export:** One-click full-database snapshot export (`.json`) or isolated active node extraction.

3.2 CMSys V2 (Tactical Monolith & Mobile PWA)

- **Path:**
- **Role:** Core Operational Command & Field Coordination System.
- **Stack:**
    - Frontend: Vanilla JavaScript (`app.js` ~2.1MB), static Tailwind (`tailwind-static.css`), HTML5 (`index.html` ~1.08MB).

    - Mobile: PWA offline-enabled frontend (`mobile.html`, `mobile.js`, `sw.js`).
    - Backend: PHP 8.x (`api.php`, `config.php`), MySQL PDO (`ncw_ps_db`), local printer routing (`get_printers.php`).
- **Key Modules:**
    - **Muster & Duty Allocation:** Morning sailor trade allocation ledger.
    - **5-Category Project Tree Accordion:**
        1. *Naval Projects* (Subdivided by 6 statutory votes: 2104, 2001, 2506, 1303, 2103, Non-Naval)
        2. *Government Projects*
        3. *Religious Projects*
        4. *Housing Projects*
        5. *Miscellaneous Projects*
    - **Tactical Job Cards & Material Logs:** Granular tasks linked to macro work orders with automated inventory decrementing.
    - **Multi-source Labour Proceed Engine:** Resolves crew assignments across memory stores, tactical allocations, and job card bindings without blocking on zero-crew planning drafts.

3.3 CMSys Logistics

- **Path:**
- **Role:** Enterprise Supply Chain & Material Procurement Engine (Port 8000).
- **Stack:** Next.js 15, React 19, TypeScript, Prisma ORM, Tailwind CSS.
- **Responsibilities:**
    - Official Naval Form **NAV 254** (Inter-Store Transfer Vouchers) and **NAV 255** (Store Inward Invoices).
    - Central Store, Zone Store, Ready-Use Store, and Balance Store tracking.
    - Reorder thresholds, stock valuation, and supplier delivery logs.

3.4 CMSys-Waterworks & CMSys-Inventory

- **CMSys-Waterworks:**
    - Modern Vite + TypeScript + Tailwind SPA dedicated to base-wide water distribution networks, pump stations, flow metrics, and pipeline maintenance.

- **CMSys-Inventory:**
  - Dedicated PHP/MySQL and Firebase migration tool (`migrate.php`) for building materials and store accounting.

4. Data Architecture & Relational Schema (`ncw_ps_db`)

The on-premises persistence layer is structured under MySQL (`ncw_ps_db`). The table specifications are detailed below:

Table Definitions Summary:

1. `users` — 667 records: Official Number, Full Name, Rank, Trade (`MA, CA, PA, PL, BB, RW, WE, AL, SW, S/S, LME, ME, OJT`), Category (`Regular`, `VSS`), Zone, Role (`Command`, `OIC`, `Supervisor`, `Sailor`), Performance Score.
2. `locations` — Naval infrastructure registry: Base, Zone (`A`, `BC`, `D`, etc.), Category (`A`–`G`), Building Name, Building No, Dimensions (Length, Breadth, Height), Stories, Structural Details (Roof/Slab), GPS Coordinates.
3. `work_orders` — Macro engineering jobs: Type (`PROJECT`, `JOB`, `TASK`), Ref No (Minute Sheet, Signal, Letter), Description, Status (`Pending`, `Active`, `Hold`, `Completed`, `Cancelled`), Priority (`High`, `Medium`, `Low`), Budget, Progress %.
4. `job_cards` — Tactical workshop tasks: Job Number (`JC/YYYY/####`), Work Order FK, Zone, Start Date, End Date, Status, Total Material Cost, Total Labor Hours.
5. `job_card_materials` — Material consumption ledger: Job Card FK, Inventory FK, Material Name, Quantity, Unit, Cost per Unit, Total Cost.
6. `job_card_labor` — Labor hours logged per sailor per date on a job card.
7. `inventory` — Stores & Materials: Category (`BMS`, `Plumbing`, `Metal`, `General`, `Aluminium`, `Paint`, `Electrical`, `Tools`), Deno (units), Quantity, Unit Cost, Location, Stock Book No, Min Stock Level.
8. `inventory_on_charge` — Inward receipts: NAV 254/255 voucher reference, Quantity, Date.
9. `daily_allocations` — Daily muster assignments: Date, Sailor FK, Work Order FK, Role, Status.
10. `daily_evaluations` — Multi-dimensional ratings (1–10 scale): Quality, Efficiency, Discipline, Material handling, Attitude, Skill, Mandatory reasons for low (1-2) or high (10) scores.
11. `estimates` & `estimate_items` — Pre-execution Bill of Quantities (BOQ) with store availability checks.

12. `sailor_board_entries` — Formal conduct board logs (White / Black Board, incident dates, penalty hours/days).
13. `sailor_performance_awards` — Weekly and monthly top trade performer recognition.
14. `user_feedback` & `feedback_tokens` — Customer/Officer sign-off on completed maintenance jobs with ratings (1–5).
15. `attendance_log` — Daily nominal roll check-in/out.
16. `zone_teams` — Permanent zone complement mappings.
17. `maintenance_records` — Periodic and preventive infrastructure maintenance records.
18. `audit_log` — System change audit logging (Table, Record ID, Action, Old JSON, New JSON, User, IP).

5. Naval Business Logic & Mathematical Invariants

5.1 Headcount Conservation Law (Zero Headcount Loss)

Every single active naval rating attached to the Civil Engineering Department must be strictly accounted for on any operational date:

$$\text{Master Complement (667)} = \sum \text{Duties} + \sum \text{Projects} + \sum \text{Leaves / Medical} + \sum \text{Camp / Yard Standby}$$

Master Complement (667) = ∑ Duties + ∑ Projects + ∑ Leaves / Medical + ∑ Camp / Yard Standby

```
Master Complement (667 Sailors)
├── Volunteer Naval Force (VAS / VSS): 577 Sailors
└── Regular Navy Cadre:                 90 Sailors
```

Enforcement Algorithm:

1. **Deduplication Quarantine:** A memory Set `allAllocatedSailorIds` is instantiated at muster runtime. No sailor can occupy multiple categories.
2. **Duty Deduplication:** Sailors assigned to active Work Orders, Job Cards, or Daily Allocations are mapped. Sailors with active leave codes are strictly blocked from duties.
3. **Detached Deployments:** Sailors allocated to detached long-term projects are gathered into the project branch allocations.
4. **Leave & Medical Register:** Remaining personnel are evaluated against `/availability/{YYYY-MM}/{DD}` for anchored status codes:
    - `LEAVE`, `L`, `TD` (Temporary Duty), `RD` (Recreation Day), `ADM` (Hospital Admitted), `S/R` (Sick Report), `SIQ` (Sick In Quarters), `NGH`, `AWOL`.

5. **Standby Default Allocation (Conservation Rule):** Any sailor who remains unassigned to duties and is not on leave automatically cascades into:
   AVAILABLE / STANDBY SAILORS (CAMP / YARD)
   AVAILABLE / STANDBY SAILORS (CAMP / YARD) This guarantees that the sum across all rows and trade columns equals exactly **577 VSS + 90 Regular = 667 sailors**.

5.2 Trade Branch Matrix

Personnel are categorized into 13 strict trade columns:

- **Volunteer Naval Force (VSS / VAS — 9 Trades):**
  `MA` (Mason), `CA` (Carpenter), `PA` (Painter), `PL` (Plumber), `BB` (Bar Bender), `RW` (Road Worker), `WE/WL` (Welder), `AL` (Aluminum Fabricator), `SW` (Signwriter).
- **Regular Navy Ratings (4 Trades):**
  `S/S` (Senior Sailors: CPO, PO, Chief), `LME` (Leading Marine Mechanic / Engineering), `ME` (Marine Mechanic / Able Rating), `OJT` (On-the-Job Trainee / Apprentice).

6. Entity Interconnection Matrix (`REL-01` ~ `REL-26`)

The following registered relationships define cross-node integrity across the ecosystem:

Rel ID	Origin Node / Table	Target Node / Table	Cardinality	Operational Description
REL-01	`work_orders`	`job_cards`	$1 : N$	Macro Work Orders partition into granular tactical Job Cards.
REL-02	`work_orders`	`estimates`	$1 : N$	Work Orders link to approved Bill of Quantities (BOQ).
REL-03	`approved_projects`	`work_orders`	$1 : N$	NHQ High-value projects cascade into zonal work orders.
REL-04	`locations`	`work_orders`	$1 : N$	Base infrastructure buildings & jetties bind to work orders.
REL-05	`locations`	`job_cards`	$1 : N$	Specific rooms, compartments, or floors bound to job cards.
REL-06	`job_cards`	`job card_materials`	$1 : N$	Tracks physical building materials consumed during execution.
REL-07	`inventory`	`job card_materials`	$1 : N$	Material consumption automatically decrements store inventory.
REL-08	`inventory`	`estimates`	$1 : N$	Catalog store items projected in pre-work cost estimates.
REL-09	`sailors`	`daily_allocations`	$1 : N$	Sailors assigned to daily duty rosters per calendar date.
REL-10	`work_orders`	`daily_allocations`	$1 : N$	Work orders staff daily assigned sailor crew details.
REL-11	`sailors`	`availability`	$1 : N$	Leave, Sick, Hospitalization, and AWOL daily status ledger.
REL-12	`inventory`	`nav254_vouchers`	$1 : N$	Inter-store transfer documented under Naval Form NAV 254.
REL-13	`inventory`	`temp_issues`	$1 : N$	Power tools & machinery loaned via Temp Issue Book.
REL-14	`sailors`	`daily_evaluations`	$1 : N$	Trade performance evaluated on 6-axis 1–10 scale.
REL-15	`work_orders`	`daily_evaluations`	$1 : N$	Evaluations linked to specific project work order context.
REL-16	`job_cards`	`user_feedback`	$1 : 1 1 : 1$	Completed job cards require client/officer rating verification.
REL-17	`sailors`	`job_card_labor`	$1 : N$	Detailed hours worked per sailor logged against job cards.

Rel ID	Origin Node / Table	Target Node / Table	Cardinality	Operational Description
REL-18	`locations`	`maintenance_records`	$1 : N$	Historical maintenance records tied to building lifecycle.
REL-19	`sailors`	`zone_teams`	$1 : 1 1 : 1$	Base organizational assignment into zone workshops.
REL-20	`sailors`	`attendance_log`	$1 : N$	Gate / Base nominal roll check-in and check-out logs.
REL-21	`sailors`	`sailor_board_entries`	$1 : N$	White Board commendations and Black Board disciplinary logs.
REL-22	`sailors`	`performance_awards`	$1 : N$	Best trade worker awards computed over weekly/monthly cycles.
REL-23	`inventory`	`inventory_on_charge`	$1 : N$	Store inward stock additions via receipt vouchers.
REL-24	`work_orders`	`naval_projects_vote`	$1 : 1 1 : 1$	Statutory Vote allocations (2104, 2001, etc.) funding works.
REL-25	`users`	`audit_log`	$1 : N$	User mutation footprint across all system entities.
REL-26	`feedback_tokens`	`job_cards`	$1 : 1 1 : 1$	Cryptographic single-use token authorizing external feedback.

7. Security, Deployment & Environmental Configuration

7.1 Security Architecture

- **API Security:** CORS origin filtering restricting requests to `http://localhost`, `http://127.0.0.1`, `http://localhost:8000`.
- **Response Headers:** `X-Frame-Options: SAMEORIGIN`, `X-Content-Type-Options: nosniff`, `Referrer-Policy: strict-origin-when-cross-origin`, `X-XSS-Protection: 1; mode=block`.
- **Input Sanitization:** Client-side HTML entity escaping (`escapeHtml()`) across modal inspectors; PDO prepared statements with strict parameter binding in `api.php`.

7.2 Deployment Pipelines (3-Tier Standard)

```
[Tier 1: Local Development]
├── Local Apache / MySQL: c:\xampp\htdocs\CMSys-DB-Hub & CMSys V2
└── Local Node / Bun:      c:\xampp\htdocs\CMSys Logistics (Port 8000)
       │
       ▼ (Git push to repository)
[Tier 2: Staging / Beta Deployment]
├── Desktop Beta: cmsys-beta.vercel.app
└── Mobile Beta:  Field-tested on naval tablets / Android supervisors
       │
       ▼ (Command approval)
[Tier 3: Production Live Base Deployment]
└── Live Production: cmsys-live.vercel.app / Base Local Server
```

8. Architectural Assessment & Recommendations

1. **Two-Way Live Sync in CMSys-DB-Hub:**
   Currently, edits made in `CMSys-DB-Hub` are persisted in memory and require manual trigger to export. A direct `PATCH` endpoint integration should be enabled to write back to the Firebase REST API securely with authentication tokens.
2. **Tokenization of Firebase Secrets:**
   Consolidate the API Keys in `config.php` and `index.html` into a protected environment file (`.env.local`) to safeguard credentials.
3. **Modularization of `CMSys V2/app.js`:**
   The ~2.1MB monolith script should gradually be refactored into ES modules matching the architecture established in `CMSys Logistics`.
4. **Automated Scheduled Backups:**
   Deploy an automated daily cron job using the existing `sync_and_seed_database.php` script to archive dated JSON and SQL snapshots into cold storage.

End of Technical Specification Document.